

ECE 501 DIGITAL IMAGE PROCESSING
SAR Optical Feature Fusion For Crop Type Classification
REPORT WEEK 1
GROUP 11

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1) Introduction and Problem Statement

The use of the crop-type mapping and yield estimation using the data from the satellite remote sensing is one of the key applications of agriculture remote sensing. In this respect, Optical sensors as sentinel-2 rely on and make use of the spectro-reflectance signature (thus sensitive to pigment and canopy-like property) while SAR data (Sentinel-1) rely on and make use of the back-scattered signal of the vegetation-canopy (sensitive to structural and dielectric-like property), and are independent on daytime or clouds. There are many apparent disadvantages for both SAR and optical (most of the Optical imagery is obscured during crop growing-season, SAR is more noisy, less directly related with biophysical variables).

In this project, “SAR-Optical Feature Fusion,” we plan to solve this problem by combining classic hand-crafted features derived from only SAR and optical data of field crops to classify the crop type and, if available, relate to yield information in the form of per-pixel yield masks. We need to develop three classical ML pipelines, i.e. pure Optical, pure SAR and multi-modal information and assess their performance with regard to classification accuracy, F1-score, and confusion matrices to show improvement gains achieved by fusion compared with two individual sources.

Finally, tasks required for this project include:

- (i) extract classical and hand-crafted features on each modality of data including sentinel-2 optical data, sentinel-1 SAR data and its associated crop-type label, per-pixel yield masks (if given);
- (ii) apply feature level fusion between these two lists of features
- (iii) train and evaluate classical ML models (e.g. Random Forest, Support Vector machine) on optical-only features, SAR-only features and fused features to report and compare their accuracy, F1 score and confusion matrices.

2) Reasons why we select this topic

A set of reasons are closely related to the objective of a Digital Image Processing course-project and this is the reason our SAR – Optical Feature Fusion topic was selected. First, it has two very different imaging mechanisms within the same workflow – passive optical multispectral imaging and coherent radar imaging, and gives us the opportunity to apply and evaluate classical DIP algorithms (spectral indices calculation, textures with GLCM, filtering/despeckling, features normalization, etc) on a true heterogeneous satellite dataset, instead of on either one modality only or synthetic data. Secondly, the problem, per se, is practically-motivated: optical images are degraded during the relevant part of the growing season, which may be critical for crop growth in certain geographic regions (tropical/humid countries, such as Brazil and Uruguay). SAR imaging can perform a real time imaging without needing to be under or near an optical sky; assessing to what extent SAR images can be used to supplement or complement the missing information obtained by optical images thus becomes a realistic question of importance in Remote Sensing. The chosen fusion in the feature level enables us to separate out the optical-only, SAR-only, and the fusion-enhanced cases, and then analyze the fusion-enhanced change.

Third, the assignment was designed without the use of a deep learning model; instead, features were hand crafted and shallow learning models were used. This fits well with an introduction level course in digital image processing, and it is our responsibility to know the physical meaning of the spectral bands, index and SAR statistics and Backscatter instead of an already learned representation. This makes it also possible to process it in a feasible way in the usual time and with the limited number of people available for a course project in a semester.

Lastly, the access to a specially-released (and thus relatively young) Benchmark dataset (YieldSAT), focusing on the specific country, crop and target type suggested in the statement of problem, was a factor in our choice making it feasible within the constraints of the term, so that time was still left for exploration in our own SAR system and related feature generation/fusion strategy.

3) Brief Background

SAR/Optical multi-sensor fusion for agricultural mapping is an ongoing research area in remote-sensing. Previous studies have shown the ability of fusing Sentinel-1 and Sentinel-2 time series in crop-type classification and monitoring of crop phenology to be superior to using either Sentinel-1 or Sentinel-2 alone, particularly in cloudy regions, and that classical machine learning models, such as Random Forest and Gradient Boosting, are competitive baseline models for crop-type mapping and phenology-related tasks when used with engineered features in the Sentinel-1 and Sentinel-2 spectra and radar-backscatter. Furthermore, several studies have combined Sentinel-1 with optical (Landsat-8/Sentinel-2) image data in agricultural regions in Brazil, where Sentinel-1

plays a key role in crop type mapping, which is a subject of relevance to the South-American country group in this assignment.

It motivates the classical (non-deep-learning) set of features to be extracted from our project: optical (e.g., NDVI, NDRE, NDMI), SAR (VV, VH, VV/VH), and concatenation of features at the feature level for the fused setting. The SAR data will not be included in the provided YieldSAT package, but will be provided by the group through Google Earth Engine (GEE) and will be for the same fields provided in the YieldSAT package. This acquisition will be done in this manner:

Specify the field boundary polygons / centroid coordinates and acquisition dates for each YieldSAT field to define the area and time of interest.

The Sentinel-1 data was pre-processed to match the dates of the growing season of the optical/yield samples for each field in the same years, using interferometric Wide (IW) swath mode (VV and VH polarizations).

Apply the typical GEE preprocessing that is possible for Sentinel-1 GRD (border noise removal, thermal noise removal, radiometric calibration to backscatter coefficient σ^0 , and terrain correction); and optionally a speckle-reduction filter (e.g., Lee or Refined Lee) prior to feature extraction.

For each field, export the SAR patch, in the same format as the optical patch geometry, and spatially co-register with the optical pixels for ease in fusion.

4) Dataset Identification

A series of optical pictures, labels of crop type and yield masks are generated and produced. The problem statement, which contains information on Sentinel-2 optical imagery availability, crop-type labels and yield masks for two out of four countries (called "YieldSAT"). We discovered that this is the recently published benchmark for crop yield prediction, which is multimodal, high resolution and is called YieldSAT (Miranda et al., CVPR 2026, project page: yieldsat.github.io).

These are the principal ones that we will be using for this project:

- The coverage is made up of 4 countries (Group A - Germany, Uruguay and Group B - Argentina, Brazil) with approx. 138,288 hectares of carefully selected fields.
- Crops: which are not associated with agriculture, but are important are primarily the crops, including corn (maize), rapeseed, soya beans and wheat.
- Yield labels: Over 12.2 million yield samples per pixel (10 m spatial) from GPS tagged yield-monitor readings from combine harvesters - wet yields, moisture content, and time-stamp by point - are then rasterized to the spatial resolution of the satellite.
- Optical imagery: 113,555 time-series image patches from the optical bands of Sentinel-2, labelled with crop and with yield masks, and Other Data: weather, soil

and topography (not used, only optical bands, crop-type labels and yield masks used).

- The data set and codes to create the benchmark are available at the official project website, and the information on the data acquisition and pre-processing can be found in the paper.

One of the key attributes of YieldSAT is that it is a yield regression dataset (as stated in the problem itself) and we will be learning a “crop-type label” per field directly from cropping attribute (corn / rapeseed / soybean / wheat) that is a 2-to-4-class classification problem that we have been assigned to our country pair.

5) Planned Methodology

- Data preparation: Download the optical/label/yield data for the two specified countries from YieldSAT.
- Retrieve and co-register co-aligned Sentinel-1 SAR data on GEE; Optical and SAR data co-registration and clipping to the same field patches.
- Optical feature extraction: For each field/patch calculate vegetation indices (NDVI, NDRE, NDMI, EVI) and GLCM texture features (contrast, homogeneity, energy, correlation).
- SAR feature extraction: Compute the mean, variance of VV, variance of VH, and variance of the ratio of VV/ VH from SAR patches for each field; Compute the GLCM texture features for each SAR patch in each field.
- Feature fusion: Normalize and combine the optical and SAR feature vectors for each field together to form a fused feature vector.
- Model training: Train the ML classifiers individually for the optical, SAR, and fused datasets, with the same cross-validation, train-test split and hyperparameter tuning for a fair comparison.

6) Work completed this week and Plan Ahead

With this being the Week 1 submission, the group has been working on understanding the problem, background information, and identifying/confirming the data sets and tools required, but not on implementation. So far, there is no code and feature extraction.

Then we will conduct a literature review on the problem statement for SAR-optical fusion and pinpoint the county of the yield-sat dataset and confirm counties assigned. next week. We are also planning to outline GEE based Sentinel-1 acquisition plan.

7) References

- [1] A. Miranda et al., “YieldSAT: A Multimodal Benchmark Dataset for High-Resolution Crop Yield Prediction,” CVPR 2026. Project page: <https://yieldsat.github.io/>
- [2] Google Earth Engine, “Sentinel-1 SAR GRD: C-band Synthetic Aperture Radar Ground Range Detected,” Data Catalog documentation, Google LLC.
- [3] N. Gorelick et al., “Google Earth Engine: Planetary-scale geospatial analysis for everyone,” Remote Sensing of Environment, 2017.
- [4] Studies on Sentinel-1/Sentinel-2 fusion for crop-type mapping and phenology estimation in Brazil and Germany (to be expanded with full citations during the Week 2 literature review).